

5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s) or (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	M1
	around a loop/around a <u>closed</u> circuit	A1
5(b)(i)	$I = 1.8 / 0.90$ $= 2.0 \text{ A}$	A1
5(b)(ii)	$Q = It$	C1
	number = $(2.0 \times 45) / 1.60 \times 10^{-19}$ $= 5.6 \times 10^{20}$	A1
5(b)(iii)	$4.0 = 1.8 + [2.0 \times (0.35 + R)]$ or $4.0 = 2.0 \times (0.90 + 0.35 + R)$	C1
	$R = 0.75 \Omega$	A1
5(c)(i)	$1.2 / 1.8 = 0.30 / L$	C1
	$L = 0.45 \text{ m}$	A1
5(c)(ii)	p.d. across XY decreases/p.d. across XP decreases	B1
	(so) P is moved towards Y/away from X/to the right	B1